

MA1521 L1 Week 03 Wed 20260826

1 variable input

$$|d\vec{r}/dt|$$

For each of the following vector functions of time, calculate the velocity, speed $|ds/dt|$, unit tangent vector (in the direction of velocity), and acceleration.

a) $e^t \mathbf{i} + e^{-t} \mathbf{j}$

b) $t^2 \mathbf{i} + t^3 \mathbf{j}$

c) $(1 - 2t^2) \mathbf{i} + t^2 \mathbf{j} + (-2 + 2t^2) \mathbf{k}$

$$\vec{r}(t) =$$

$$\vec{r}(t) =$$

$$\vec{r}(t) =$$

"Position" $\vec{r}(t)$

a) $\begin{pmatrix} e^t \\ e^{-t} \end{pmatrix}$

b) $\begin{pmatrix} t^2 \\ t^3 \end{pmatrix}$

c) $\begin{pmatrix} 1 - 2t^2 \\ t^2 \\ -2 + 2t^2 \end{pmatrix}$

Velocity $\frac{d\vec{r}}{dt}(t)$

a) $\begin{pmatrix} e^t \\ -e^{-t} \end{pmatrix}$

b) $\begin{pmatrix} 2t \\ 3t^2 \end{pmatrix}$

c) $\begin{pmatrix} -4t \\ 2t \\ 4t \end{pmatrix}$

Speed $\left| \frac{d\vec{r}}{dt}(t) \right|$

a) $\sqrt{e^{2t} + e^{-2t}}$

b) $\sqrt{4t^2 + 9t^4}$

c) $\sqrt{16t^2 + 4t^2 + 16t^2} = 6t$

Unit tangent vector $\frac{\frac{d\vec{r}}{dt}(t)}{\left| \frac{d\vec{r}}{dt}(t) \right|}$

a) $\frac{\begin{pmatrix} e^t \\ -e^{-t} \end{pmatrix}}{\sqrt{e^{2t} + e^{-2t}}}$

b) $\frac{\begin{pmatrix} 2t \\ 3t^2 \end{pmatrix}}{\sqrt{4t^2 + 9t^4}}$

c) $\frac{1}{6t} \begin{pmatrix} -4t \\ 2t \\ 4t \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$

Acceleration $\frac{d^2\vec{r}}{dt^2}(t)$

a) $\begin{pmatrix} e^t \\ e^{-t} \end{pmatrix}$

b) $\begin{pmatrix} 2 \\ 6t \end{pmatrix}$

c) $\begin{pmatrix} -4 \\ 2 \\ 4 \end{pmatrix}$

of t

Prove the rule for differentiating the scalar product of two plane vector functions:

$$\frac{d}{dt} \mathbf{r} \cdot \mathbf{s} = \frac{d\mathbf{r}}{dt} \cdot \mathbf{s} + \mathbf{r} \cdot \frac{d\mathbf{s}}{dt}, \quad \checkmark$$

by calculating with components, letting $\mathbf{r} = x_1 \mathbf{i} + y_1 \mathbf{j}$ and $\mathbf{s} = x_2 \mathbf{i} + y_2 \mathbf{j}$.

$$\approx \begin{pmatrix} x_1(t) \\ y_1(t) \end{pmatrix} \quad \approx \begin{pmatrix} x_2(t) \\ y_2(t) \end{pmatrix}$$

$$(\mathbf{r} \cdot \mathbf{s})(t) = \mathbf{r}(t) \cdot \mathbf{s}(t) = x_1(t) x_2(t) + y_1(t) y_2(t)$$

$$\frac{d}{dt}(\mathbf{r} \cdot \mathbf{s})(t) = x_1'(t) x_2(t) + x_1(t) x_2'(t) + y_1'(t) y_2(t) + y_1(t) y_2'(t) \quad (*)$$

$$\frac{d\mathbf{r}}{dt}(t) = \begin{pmatrix} x_1'(t) \\ y_1'(t) \end{pmatrix}, \quad \left(\frac{d\mathbf{r}}{dt} \cdot \mathbf{s} \right)(t) = x_1'(t) x_2(t) + y_1'(t) y_2(t)$$

$$\frac{d\mathbf{s}}{dt}(t) = \begin{pmatrix} x_2'(t) \\ y_2'(t) \end{pmatrix}, \quad \left(\mathbf{r} \cdot \frac{d\mathbf{s}}{dt} \right)(t) = x_1(t) x_2'(t) + y_1(t) y_2'(t)$$

these
add up
to give
(*)

Suppose a point P moves on the surface of a sphere with center at the origin; let
 $OP = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$.

Show that the velocity vector $\mathbf{v}$ is always perpendicular to $\mathbf{r}$

Prove the converse: if $\mathbf{r}$ and $\mathbf{v}$ are perpendicular, then the motion of P is on the surface of a sphere.

$\vec{OP} = \mathbf{r}(t)$ lies on sphere center at the origin say of radius $a > 0$

i.e. $|\mathbf{r}(t)| = a$, is const, indep of t

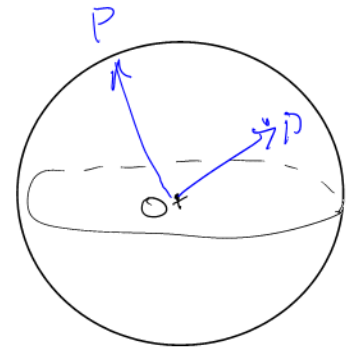
i.e. $|\mathbf{r}(t)|^2 = \mathbf{r}(t) \cdot \mathbf{r}(t) = a^2$, indep of t

so its derivative is 0, i.e. $\frac{d}{dt} \mathbf{r} \cdot \mathbf{r}(t) = 0$

By prev. qn, $\frac{d\mathbf{r}}{dt} \cdot \mathbf{r} + \mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$, i.e. $\frac{d\mathbf{r}}{dt}(t) \cdot \mathbf{r}(t) = 0$

hence $\mathbf{v}(t)$ always $\perp$ to $\mathbf{r}(t)$

velocity vector $\mathbf{v}(t)$



converse : Assume P moves such that $\mathbf{v}(t)$ always $\perp$ to $\mathbf{r}(t) = \vec{OP}(t)$

WTS: P always on a sphere (centered at O)

If. Assumption says $\underbrace{\frac{d\mathbf{r}}{dt}(t)}_{\mathbf{v}(t)} \cdot \mathbf{r}(t) = 0$, so $\frac{d}{dt} \mathbf{r} \cdot \mathbf{r}(t) = 0$

i.e., $\mathbf{r}(t) \cdot \mathbf{r}(t)$ is a const. ≥ 0
say a^2

i.e., $|\mathbf{r}(t)|^2 = a^2$ const, indep of t

i.e., $|\mathbf{r}(t)| = a$, const, indep of t
 $\vec{OP}(t)$

i.e., P always on a sphere, radius a , centered at O

a) Prove the formula $\frac{d}{dt} u(t) \mathbf{r}(t) = \underbrace{\frac{du}{dt}}_{\text{scalar fn}} \mathbf{r}(t) + u(t) \underbrace{\frac{d\mathbf{r}}{dt}}_{\text{vector fn}}$.

(You may assume the vectors are in the plane; calculate with the components.)

b) Let $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j}$, the exponential spiral. Use part (a) to find the speed of this motion.

a) Suppose $\mathbf{r}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ Then $u(t) \mathbf{r}(t) = \begin{pmatrix} u(t) x(t) \\ u(t) y(t) \end{pmatrix}$

Then $\frac{d}{dt} u(t) \mathbf{r}(t) = \begin{pmatrix} u'(t) x(t) + u(t) x'(t) \\ u'(t) y(t) + u(t) y'(t) \end{pmatrix}$

$$= \begin{pmatrix} u'(t) x(t) \\ u'(t) y(t) \end{pmatrix} + \begin{pmatrix} u(t) x'(t) \\ u(t) y'(t) \end{pmatrix}$$

$$= u'(t) \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} + u(t) \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix} = u'(t) \mathbf{r}(t) + u(t) \mathbf{r}'(t)$$

$$b) \quad \mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} = e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

Using (a)
Vel. vector $\mathbf{r}'(t) = \frac{d}{dt} e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + e^t \frac{d}{dt} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$

$$= e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + e^t \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} = e^t \begin{pmatrix} \cos t - \sin t \\ \sin t + \cos t \end{pmatrix}$$

so speed is $|\mathbf{r}'(t)|$ (as fn of t)

$$= |e^t| \cdot \left| \begin{pmatrix} \cos t - \sin t \\ \sin t + \cos t \end{pmatrix} \right|$$

$$= e^t \cdot \sqrt{\cos^2 t - 2 \sin t \cos t + \sin^2 t + \cos^2 t + 2 \sin t \cos t + \sin^2 t} = \sqrt{2} e^t$$